

GED® Math — Formula Cheat Sheet

Mathematical Reasoning · 46 questions · 115 minutes · pass = 145/200 · certpathpublishing.store

NUMBER SENSE & PERCENTS

Fractions, Percents, Ratios

$$x\% = x/100 \quad \cdot \quad \text{"of"} = \times$$

- **% change** = (new – old) / old × 100
- **Part** = percent × whole; **Whole** = part / percent
- **Proportion**: $a/b = c/d \rightarrow$ cross-multiply ($ad = bc$)
- **Unit rate**: divide to get "per 1" (miles/hour, \$/item)
- Order of operations: **PEMDAS**

Exponents & Roots

$$a^m \cdot a^n = a^{m+n} \quad a^m/a^n = a^{m-n}$$

- $(a^m)^n = a^{mn}$ · $a^0 = 1$ · $a^{-n} = 1/a^n$
- $\sqrt{\quad}$ undoes squaring: $\sqrt{25} = 5$
- **Scientific notation**: $a \times 10^n$, with $1 \leq a < 10$
- Negative × negative = positive; neg × pos = negative

ALGEBRA & COORDINATE GEOMETRY

Lines & Slope

$$m = (y_2 - y_1) / (x_2 - x_1)$$

- **Slope-intercept**: $y = mx + b$ ($b = y$ -intercept)
- **Point-slope**: $y - y_1 = m(x - x_1)$
- **Standard form**: $Ax + By = C$
- **Parallel** = same slope; **Perpendicular** = negative reciprocal (flip & change sign)

Quadratics & Factoring

$$x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$$

- **Difference of squares**: $a^2 - b^2 = (a + b)(a - b)$
- **FOIL**: $(x + p)(x + q) = x^2 + (p+q)x + pq$
- If $x^2 = k$ then $x = \pm\sqrt{k}$
- **Distance**: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- **Midpoint**: $((x_1 + x_2)/2, (y_1 + y_2)/2)$

GEOMETRY & MEASUREMENT

Area & Perimeter

- **Rectangle**: $A = lw$ · $P = 2(l + w)$
- **Triangle**: $A = \frac{1}{2}bh$ · angles sum to 180°
- **Parallelogram**: $A = bh$
- **Trapezoid**: $A = \frac{1}{2}(b_1 + b_2)h$
- **Circle**: $A = \pi r^2$ · $C = 2\pi r$ ($\pi \approx 3.14$)
- **Pythagorean**: $a^2 + b^2 = c^2$

Volume

- **Rectangular box**: $V = lwh$
- **Cylinder**: $V = \pi r^2 h$
- **Cone**: $V = \frac{1}{3}\pi r^2 h$
- **Pyramid**: $V = \frac{1}{3}(\text{base area})h$
- **Sphere**: $V = \frac{4}{3}\pi r^3$
- **Surface area**: add the area of every face

DATA, PROBABILITY & MONEY

Statistics

$$\text{Mean} = \text{sum} \div \text{count}$$

- **Median:** middle of sorted data (average the two middle if even count)
- **Mode:** most frequent value · **Range:** max – min

Probability & Interest

$$P = \text{favorable} \div \text{total}$$

- Probability is always between **0 and 1**
- **P(not A)** = $1 - P(A)$
- **Simple interest:** $I = prt$ (principal $\times$ rate $\times$ time)

Test-day facts & strategy. The Mathematical Reasoning test has **46 questions in 115 minutes** (~2.5 min each). **Part 1** (~5 questions) is **no calculator** and comes first; **Part 2** (~41 questions) allows the on-screen **TI-30XS MultiView** calculator. You need **145 out of 200 to pass** — about 60–65% of the points, roughly 30 questions right. The GED gives you an on-screen formula sheet, but knowing the common formulas above **cold** saves the minutes you need. Lock in the easy points first (fractions, percents, area/perimeter, linear equations), then come back for systems, quadratics, and functions.

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